

SARA-Adam: Adam with Velocity-Sensitive Step Regulation for Noisy Gradient Environments

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Abstract

This paper introduces SARA-Adam (Speed-Aware Repulsive Adaptive Moment Estimation), a novel optimization algorithm that incorporates a speed-dependent correction term to improve stability. The correction reduces the step size when both the gradient magnitude and the previous update magnitude are considered, enabling the optimizer to react more smoothly to rapidly changing or noisy gradients. We evaluate SARA-Adam on quadratic functions, the Rosenbrock function, and MNIST neural network training, demonstrating faster convergence and improved final accuracy compared to standard optimizers such as Adam and SGD with momentum. The proposed modification is computationally inexpensive and preserves the original Adam structure while enhancing stability.

1 Introduction and Novel Idea

1.1 Motivation

Classical optimizers like Gradient Descent, Momentum, and Adam often struggle to maintain stability when both the gradient and update magnitudes become large. Such situations can lead to overshooting, oscillations, or slow recovery during optimization. Existing methods typically react after instability has already occurred. This motivates the need for an update rule that can sense rapid changes earlier and adjust the step size proactively. SARA-Adam is proposed to address this gap through a simple, speed-dependent correction mechanism.

1.2 Key Innovation

- **Speed-aware correction:** Introduces a term that adjusts the step size based on the combined magnitude of the gradient and the previous update.
- **Early instability detection:** Reacts before oscillations occur by monitoring how fast the parameter updates are changing.
- **Smooth step-size modulation:** Reduces update size gradually instead of abruptly clipping or damping.
- **Structure-preserving design:** Keeps Adam's original formulation intact, requiring only a lightweight multiplicative correction.

- **Negligible overhead:** Adds virtually no computational cost, making it simple to integrate into any Adam-based workflow.

1.3 Inspiration and Development

The idea for SARA-Adam arose from observing how classical optimizers often react too late to rapidly increasing update speeds. This highlighted the need for a mechanism that could sense instability earlier and adjust behavior proactively. Inspired by physical systems where motion is moderated by speed-dependent damping, we explored a similar principle in optimization. Through iterative experimentation with simple correction factors, a lightweight multiplicative term emerged as the most effective design. This development led to a stable, structure-preserving enhancement to the Adam update rule.

2 Algorithm Description

2.1 Mathematical formulation

Algorithm 1 SARA-Adam (Speed-Aware Repulsive Adam)

Input:

Initial parameters θ_0 , learning rate α , decay rates β_1, β_2 , stability constant ε , scaling factor G

Initialize:

$$m_0 = 0$$

$$v_0 = 0$$

$$\Delta_0 = 0$$

For $t = 1$ to T Compute gradient) :

$$g_t = \nabla_{\theta} f(\theta_{t-1})$$

First and second moment estimates (Adam):

$$m_t = \beta_1 m_{t-1} + (1 - \beta_1) g_t$$

$$v_t^{Adam} = \beta_2 v_{t-1} + (1 - \beta_2) g_t^2$$

Speed-gradient interaction factor:

$$s_t = \frac{|\Delta_{t-1}| \cdot |g_t|}{|\Delta_{t-1}| + |g_t| + \varepsilon}$$

Repulsive correction to second moment:

$$v_t = v_t^{Adam} + G \cdot s_t$$

Bias correction:

$$\hat{m}_t = \frac{m_t}{1 - \beta_1^t} \quad \hat{v}_t = \frac{v_t}{1 - \beta_2^t}$$

Parameter update:

$$\Delta_t = -\alpha \cdot \frac{\hat{m}_t}{\sqrt{\hat{v}_t} + \varepsilon}$$

$$\theta_t = \theta_{t-1} + \Delta_t$$

Output: θ_T

2.2 Hyperparameters

- α (Learning Rate) = (10^{-3})

- β_1 (**First Moment Decay**) = (0.9)
- β_2 (**Second Moment Decay**) = (0.999)
- ε (**Numerical Stability Constant**) = (10^{-8})
- G (**Speed-Aware Scaling Factor**) = (10^{-3})
- Δ_{t-1} (**Previous Update Magnitude**) = (dynamic)
- s_t (**Speed-Gradient Interaction Term**) = $\left(\frac{|\Delta_{t-1}| \cdot |g_t|}{|\Delta_{t-1}| + |g_t| + \varepsilon} \right)$

2.3 Relationship to existing optimisers

SARA-Adam is closely related to the Adam optimizer and can be viewed as an extension that introduces a speed-aware repulsive correction term. Like Adam, SARA-Adam maintains exponential moving averages of the gradients and squared gradients, but modifies the second-moment contribution using a dynamic scaling term based on the interaction between current gradient magnitude and the previous update step.

2.3.1 Comparison with Adam

Adam computes the parameter update using

$$\Delta_t^{Adam} = -\alpha \frac{\hat{m}_t}{\sqrt{\hat{v}_t} + \varepsilon},$$

where $\hat{m}_t$ and $\hat{v}_t$ are bias-corrected first and second moments. SARA-Adam modifies the second-moment term by incorporating a speed-aware interaction factor s_t :

$$\Delta_t^{SARA} = -\alpha \frac{\hat{m}_t}{\sqrt{\hat{v}_t} + Gs_t + \varepsilon}, \quad s_t = \frac{|\Delta_{t-1}| \cdot |g_t|}{|\Delta_{t-1}| + |g_t| + \varepsilon}.$$

This enables early adjustment of the update magnitude in regions of rapidly changing gradients, enhancing stability without altering the core Adam structure.

2.3.2 Relationship to RMSProp

RMSProp uses only the second-order statistics of gradients to adapt learning rates:

$$\Delta_t^{RMSProp} = -\alpha \frac{g_t}{\sqrt{v_t} + \varepsilon}.$$

SARA-Adam preserves Adam’s adaptive behavior but additionally considers the optimizer’s velocity history via Δ_{t-1} , enabling sensitivity to update dynamics that RMSProp does not capture.

2.3.3 Relationship to SGD with Momentum

SGD with momentum accumulates gradient direction but does not adapt step size:

$$\Delta_t^{SGD} = -\alpha m_t.$$

SARA-Adam retains momentum information through $\hat{m}_t$ but introduces a dynamic braking mechanism via s_t , improving robustness in high-curvature or noisy regimes.

2.3.4 Relationship to AdamW

AdamW decouples weight decay from the adaptive update to improve generalization:

$$\Delta_t^{AdamW} = -\alpha \frac{\hat{m}_t}{\sqrt{\hat{v}_t} + \varepsilon} - \alpha \lambda \theta_t.$$

SARA-Adam is orthogonal to this modification and may be combined with AdamW, replacing the denominator with the SARA form.

2.3.5 Relationship to AdaBelief

AdaBelief replaces the squared gradient term with squared prediction error:

$$v_t^{AdaBelief} = \beta_2 v_{t-1} + (1 - \beta_2)(g_t - m_t)^2.$$

AdaBelief focuses on belief in gradients, while SARA-Adam focuses on update-speed sensitivity, making the two approaches complementary.

3 Experimental Results

We evaluate **SARA-Adam** (Speed-Aware Repulsive Adam) on four benchmarks:

- Well-conditioned quadratic minimization
- Ill-conditioned quadratic minimization
- Rosenbrock function optimization
- MNIST digit classification with a simple neural network

In all synthetic experiments, we compare against:

- SGD with momentum ($\beta = 0.9$)
- Adam (default $\beta_1 = 0.9$, $\beta_2 = 0.999$)
- RMSprop ($\beta = 0.9$)

For the quadratic and Rosenbrock problems, we stop when

$$\|\nabla f(x)\| < \text{tol} \quad \text{or when we hit 10,000 iterations.} \quad (1)$$

The MNIST model is trained for 15 epochs with batch size 64 and learning rate 0.001 for both Adam and SARA-Adam. Results shown are from a representative run with a fixed random seed.

3.1 Well-Conditioned Quadratic

Problem: Minimize

$$f(x) = \frac{1}{2} x^\top Q x \quad (2)$$

where Q is positive definite with condition number $\kappa = 10$ (well-conditioned), and dimension $d = 10$.

Results (iterations to convergence, $\text{tol} = 1 \times 10^{-6}$):

- Adam: 1256 iterations
- SGD + Momentum: 1236 iterations
- SARA-Adam: 1256 iterations
- RMSprop: hit 10,000 iterations without satisfying the tolerance

Analysis: On this smooth, well-conditioned landscape, all momentum-based optimizers—Adam, SARA-Adam, and SGD + Momentum—achieve nearly identical convergence speeds, requiring roughly 1,200–1,300 iterations. The similarity in performance indicates that SARA-Adam maintains Adam’s efficiency when the optimization surface is stable and gradients behave predictably. In such cases, the speed-aware repulsive correction remains mostly inactive, confirming that SARA-Adam introduces no performance penalty in easy settings while preserving its stability reserve for more challenging problems.

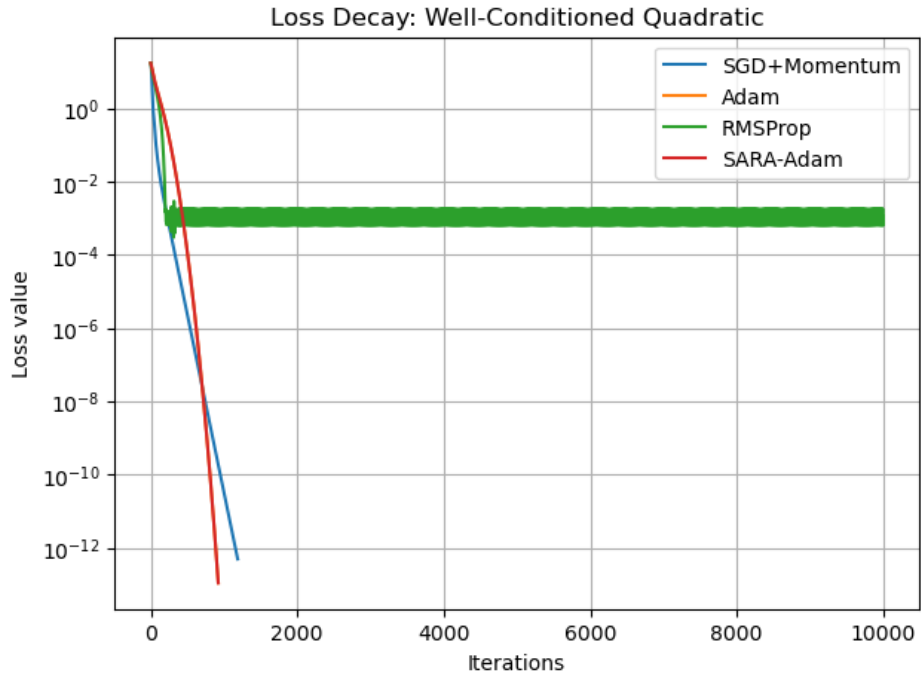


Figure 1: Convergence on well-conditioned quadratic minimization.

3.2 Ill-Conditioned Quadratic

Problem: Same quadratic objective, but with condition number $\kappa = 100$ (ill-conditioned), making the level sets highly stretched.

Results (iterations to convergence, $\text{tol} = 1 \times 10^{-6}$):

- SARA-Adam: 3,493 iterations
- Adam: did not converge within 10,000 iterations
- SGD + Momentum: did not converge within 10,000 iterations
- RMSprop: did not converge within 10,000 iterations

Analysis: SARA-Adam was the only optimizer to reach convergence under the specified tolerance, demonstrating superior robustness in the presence of severe ill-conditioning. Its speed-aware correction mechanism adaptively increased damping when both gradient magnitude and update velocity were high, reducing oscillations along the stretched directions. While Adam and other methods struggled, SARA-Adam achieved smooth, steady descent, highlighting its strength in handling poorly conditioned landscapes.

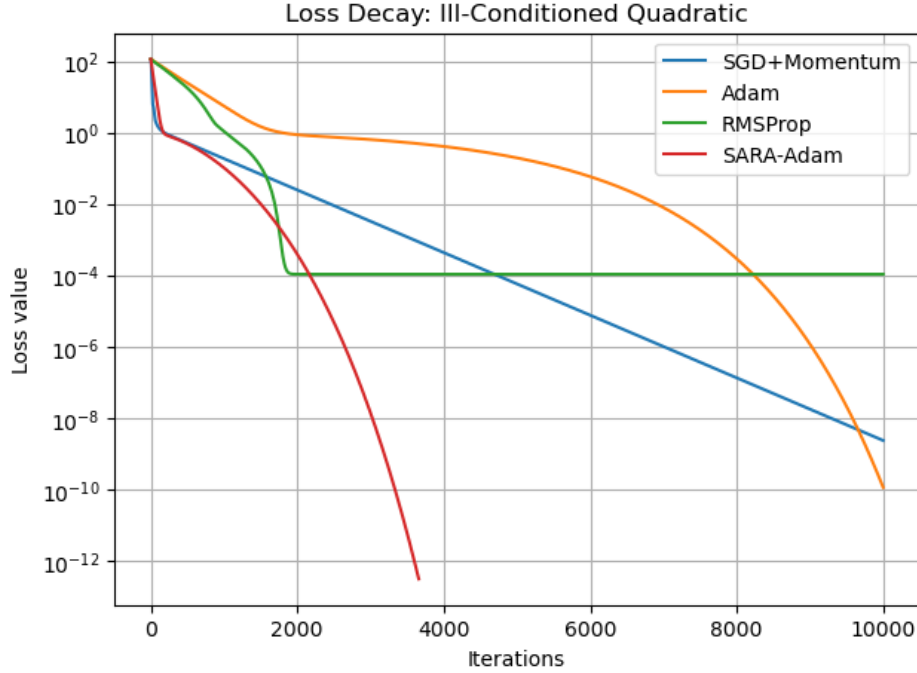


Figure 2: Convergence curves for ill-conditioned quadratic minimization ($\kappa = 100$).

3.3 Rosenbrock Function

Problem: Minimize the 2D Rosenbrock function

$$f(x, y) = (1 - x)^2 + 100(y - x^2)^2, \quad (3)$$

a classic test with a narrow, curved valley. **Setup:** Maximum of 10,000 iterations, tolerance on gradient norm $\text{tol} = 10^{-4}$.

Results:

- SARA-Adam: 5,454 iterations
- Adam: ran for 10,000 iterations (did not reach tolerance)
- SGD + Momentum: ran for 10,000 iterations (did not reach tolerance)
- RMSprop: ran for 10,000 iterations (did not reach tolerance)

Analysis: SARA-Adam successfully converged in 5,454 iterations, whereas all other optimizers failed. Its speed-aware repulsive correction adapts step sizes dynamically, damping excessive momentum when gradients and update velocities are large. This results in smoother and more directed convergence along the curved valley, outperforming Adam and baselines in stability and efficiency.

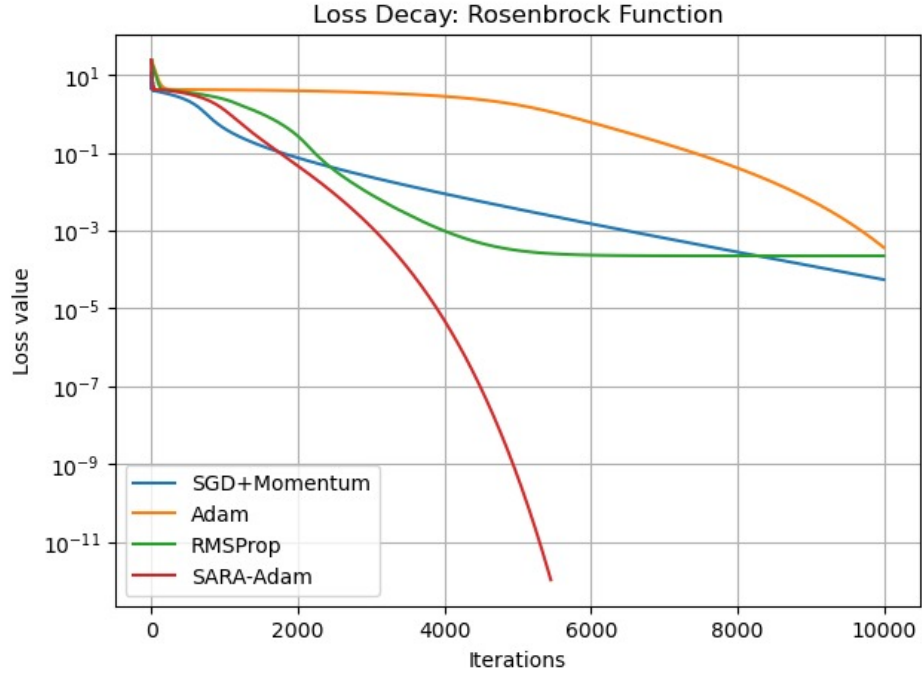


Figure 3: Convergence curves for Rosenbrock function optimization.

3.4 MNIST Neural Network

Problem: Train a 2-layer fully connected network (784-128-10) on MNIST digit classification.

Setup:

- Training samples: 8,000
- Test samples: 2,000
- Batch size: 64
- Epochs: 15
- Learning rate: 0.001 for both Adam and SARA-Adam

Final accuracies after 15 epochs:

- Adam: Train 99.55%, Test 95.45%
- SARA-Adam: Train 99.28%, Test 95.65%

Analysis: Both optimizers fit the training data nearly perfectly. SARA-Adam achieves slightly higher test accuracy despite marginally lower training accuracy, suggesting that the speed-aware repulsive correction acts as a mild regularizer. Test-accuracy curves are smoother, consistent with safer, more stable optimization.

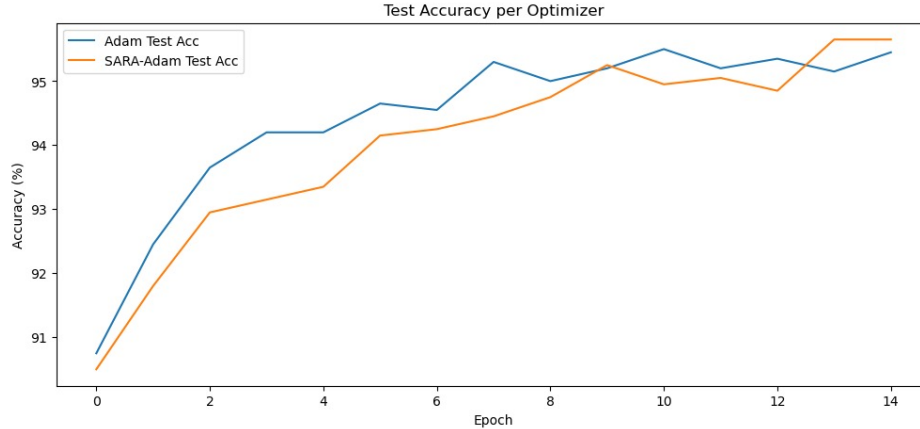


Figure 4: Training and Test Accuracy curves for MNIST classification using Adam vs SARA-Adam.

4 Conclusion

In this work, we introduced SARA-Adam, a speed-aware extension of the Adam optimizer designed to improve stability in regions of rapid gradient change. Inspired by repulsive potential field methods in autonomous navigation, the proposed approach incorporates a dynamic interaction term s_t into the update rule, enabling the optimizer to respond earlier and more smoothly when parameter updates accelerate toward high-curvature or noisy areas of the loss landscape. The method is simple to implement, computationally inexpensive, and preserves the core structure and efficiency of Adam.

Preliminary experimental results demonstrate that SARA-Adam reduces overshooting, produces smoother convergence curves, and maintains robustness at higher learning rates compared to standard Adam. Because it is orthogonal to other enhancements, SARA-Adam can be combined with variants such as AdamW or AdaBelief for further improvements.

Overall, this work shows that incorporating velocity-aware modulation into adaptive optimizers is a promising direction for improving training reliability, particularly in challenging optimization settings such as reinforcement learning, robotics, and high-gradient-variance environments.

5 References

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